

Task 1 — Iris Flower Classification

Description

Iris dataset par do machine learning models train kiye gaye hain — **K-Nearest Neighbors** aur **Random Forest** — taake phoolon ki species predict ki ja sake based on petal/sepal measurements.

Features

- Exploratory Data Analysis (EDA) with histograms aur pairplots
- Data preprocessing with `StandardScaler`
- Model training: KNN aur Random Forest
- Confusion matrix visualization
- Feature importance chart

Models Used

Model	Description
K-Nearest Neighbors	<code>n_neighbors=5</code>
Random Forest	<code>n_estimators=100</code>

Output Files

- `iris_eda.png` — Feature distributions
- `iris_pairplot.png` — Species pairplot
- `iris_confusion_matrix.png` — Model comparison
- `iris_feature_importance.png` — RF feature importance

Quick Start

```
bash
pip install numpy pandas matplotlib seaborn scikit-learn
python task1_iris.py
```

Output

Class root resolved to: /content/dataset/NCT-CRC-HE-100K

9 tissue classes found:

```
ADI      : 10,407 images
BACK     : 10,566 images
DEB      : 11,512 images
LYM      : 11,557 images
MUC      : 8,896 images
MUS      : 13,536 images
NORM     : 8,763 images
STR      : 10,446 images
TUM      : 14,317 images
```

Feature extraction shuru...

```
ADI      (200 samples)
BACK     (200 samples)
DEB      (200 samples)
LYM      (200 samples)
MUC      (200 samples)
MUS      (200 samples)
NORM     (200 samples)
STR      (200 samples)
TUM      (200 samples)
```

Feature matrix ready: (1800, 21) (samples x features)

=== Dataset Info ===

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	\
0	5.1	3.5	1.4	0.2	
1	4.9	3.0	1.4	0.2	
2	4.7	3.2	1.3	0.2	
3	4.6	3.1	1.5	0.2	
4	5.0	3.6	1.4	0.2	

```
species
0 setosa
1 setosa
2 setosa
3 setosa
4 setosa
```

Shape: (150, 5)

Class Distribution:

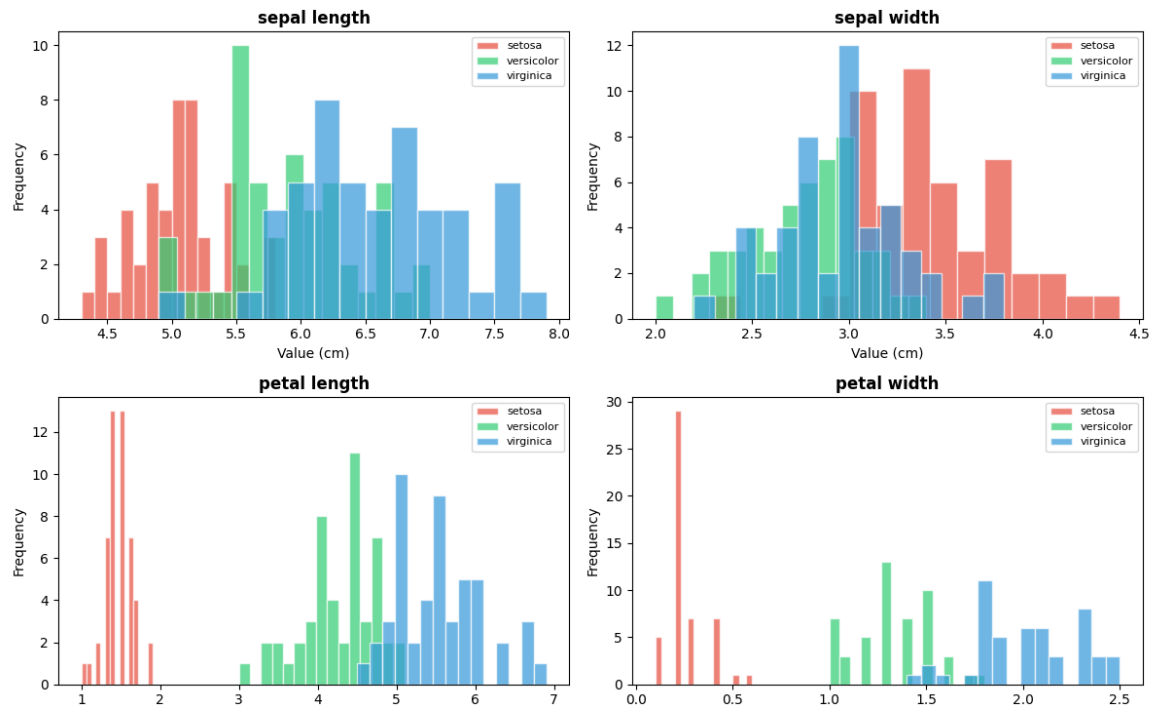
```
species
setosa      50
versicolor  50
virginica   50
Name: count, dtype: int64
```

Statistics:

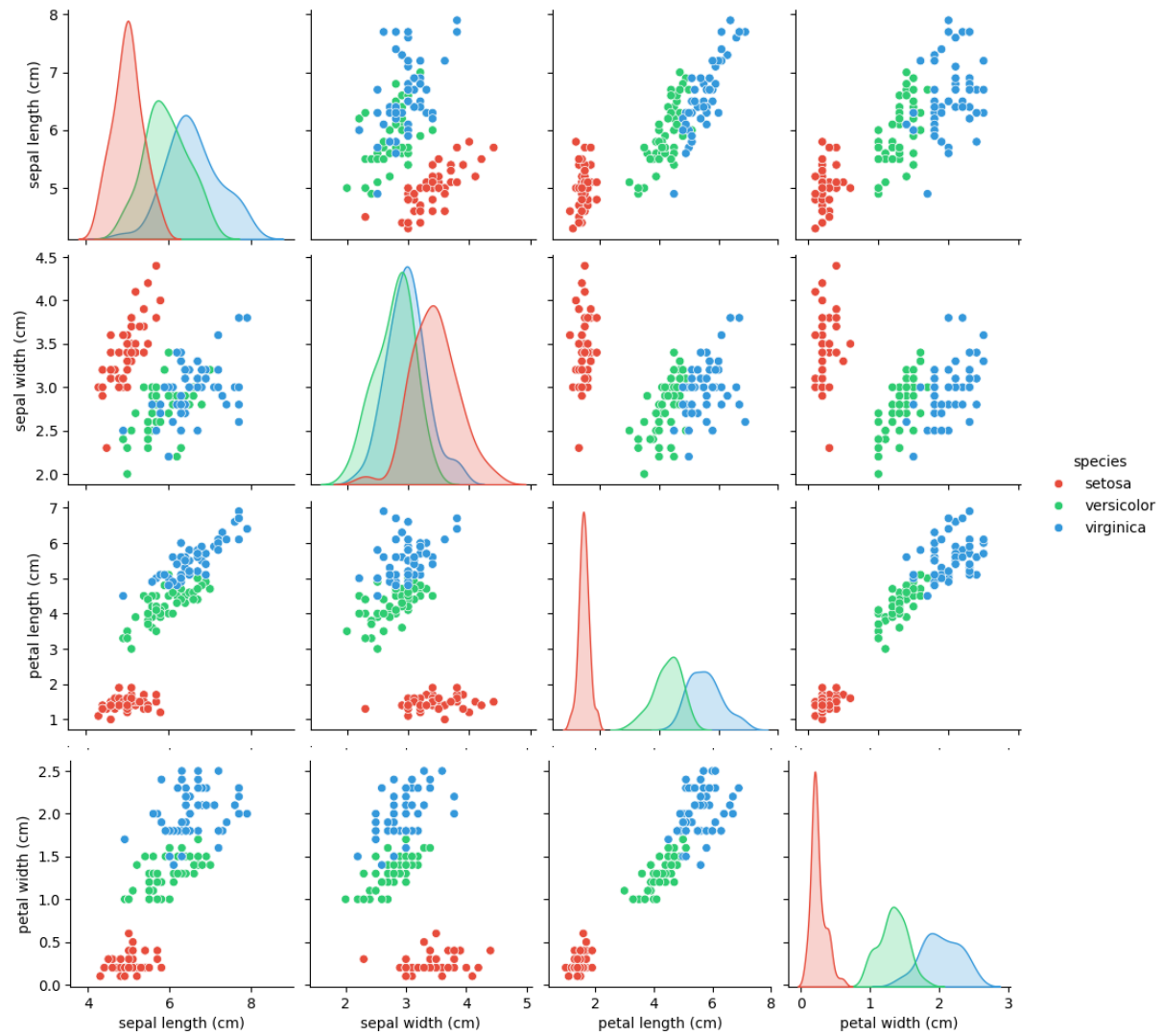
	sepal length (cm)	sepal width (cm)	petal length (cm)	\
count	150.000000	150.000000	150.000000	
mean	5.843333	3.057333	3.758000	
std	0.828066	0.435866	1.765298	
min	4.300000	2.000000	1.000000	
25%	5.100000	2.800000	1.600000	
50%	5.800000	3.000000	4.350000	
75%	6.400000	3.300000	5.100000	
max	7.900000	4.400000	6.900000	

	petal width (cm)
count	150.000000
mean	1.199333
std	0.762238
min	0.100000
25%	0.300000
50%	1.300000
75%	1.800000
max	2.500000

Iris Dataset - EDA



Iris Pairplot



=== K-Nearest Neighbors ===

Accuracy: 93.33%

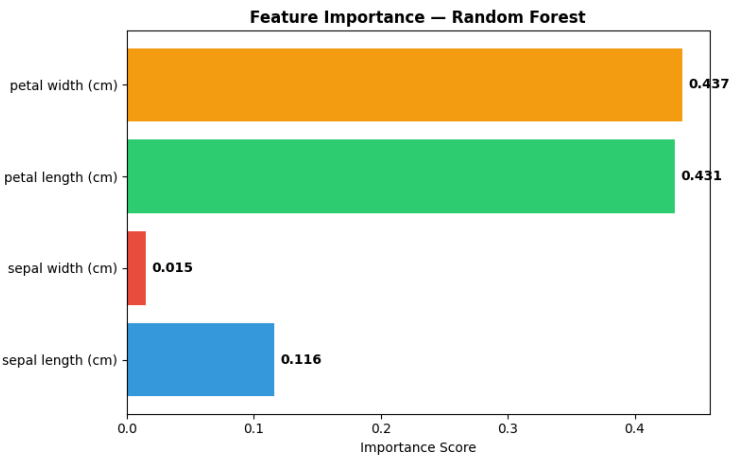
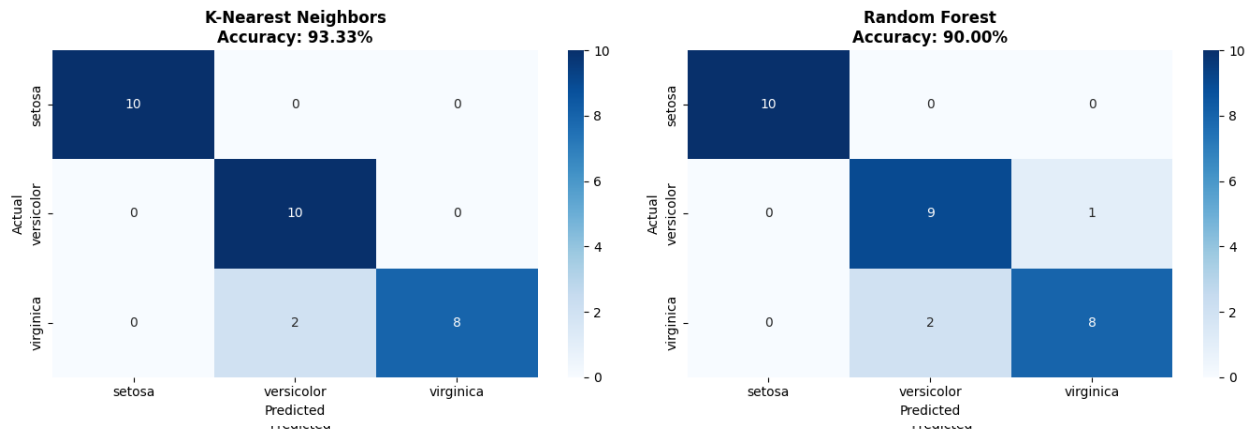
	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	10
versicolor	0.83	1.00	0.91	10
virginica	1.00	0.80	0.89	10
accuracy			0.93	30
macro avg	0.94	0.93	0.93	30
weighted avg	0.94	0.93	0.93	30

=== Random Forest ===

Accuracy: 90.00%

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	10
versicolor	0.82	0.90	0.86	10
virginica	0.89	0.80	0.84	10
accuracy			0.90	30
macro avg	0.90	0.90	0.90	30
weighted avg	0.90	0.90	0.90	30

Confusion Matrices



✔ Sample prediction: setosa